وزارة الاتصالات
وتكنولوجيا المعلومات

Mini SIEM:

Automated Security Event Detection and Analysis

Group 6

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project planning and management

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1 Project Scope & Management Objectives

1.1 Project Scope Definition

The scope defines the boundaries of the Mini SIEM project to ensure all team members and stakeholders have a clear understanding of the deliverables.

Category	In-Scope (Included)	Out-of-Scope (Excluded)
Data Types	Analysis of IPv4/IPv6, Domains/URLs, and File Hashes (MD5, SHA-1, SHA-256).	Analysis of raw network traffic packets (PCAP) or full log files.
Integrations	VirusTotal, AbuseIPDB, and AlienVault OTX APIs.	Integration with enterprise tools like Splunk, QRadar, or Sentinel.
Analysis	Rule-based detection and ML-assisted confidence scoring.	Automated remediation (e.g., automatically updating firewall rules).
User Interface	Web-based dashboard with threat visualization and query history.	Mobile application or command-line interface (CLI) for end-users.
Education	Glossary, help guides, and real-world examples for students.	Professional certification or accredited training modules.

1.2 Management Objectives

The primary management goal is to deliver a functional, secure, and well-documented prototype within the DEPI timeline.

1. **Timeline Adherence:** Complete all 6 development milestones (Ingestion to Documentation) by the final submission deadline in May 2026.
2. **Modular Development:** Ensure the codebase is 100% modular, where each API connector and engine layer is independently testable.
3. **Security Compliance:** Protect all sensitive API keys using environment variables and enforce input sanitization to prevent injection attacks.
4. **Collaborative Engineering:** Utilize GitHub for version control to manage contributions from all 5 team members effectively.

1.3 High-Level Deliverables

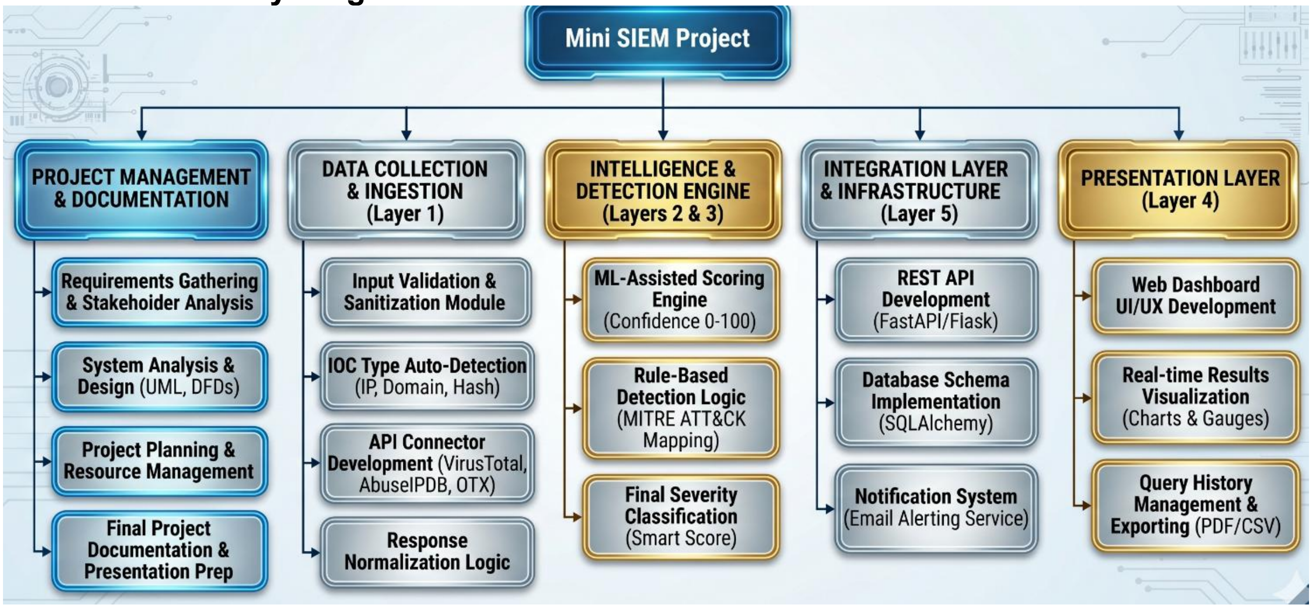
- **D1:** Technical Specification Documents (Requirements, Analysis & Design, Project Plan).
- **D2:** Functional Mini SIEM Web Application (Frontend + Backend).
- **D3:** Normalized Threat Intelligence Database (SQLite/PostgreSQL).
- **D4:** Final Project Presentation and Video Demonstration.

2. Work Breakdown Structure (WBS)

2.1 WBS Overview

The Work Breakdown Structure (WBS) decomposes the Mini SIEM project into manageable work packages. This hierarchical structure ensures that every functional and non-functional requirement is assigned to a specific development phase, allowing for better tracking and resource allocation.

2.2 WBS Hierarchy Diagram



2.3 Detailed Work Packages

The following list provides a detailed breakdown of the tasks illustrated in the hierarchy above:

- **1. Project Management & Documentation**
  - Requirements Gathering & Stakeholder Analysis.
  - System Analysis & Design (UML Diagrams, DFDs).
  - Project Planning & Resource Management.
  - Final Project Documentation & Presentation Prep.
- **2. Data Collection & Ingestion (Layer 1)**
  - Input Validation & Sanitization Module.
  - IOC Type Auto-Detection (IP, Domain, Hash).
  - API Connector Development (VirusTotal, AbuseIPDB, OTX).
  - Response Normalization Logic.
- **3. Intelligence & Detection Engine (Layers 2 & 3)**
  - ML-Assisted Scoring Engine (Confidence 0-100).
  - Rule-Based Detection Logic (MITRE ATT&CK Mapping).
  - Final Severity Classification (Smart Score).
- **4. Integration Layer & Infrastructure (Layer 5)**
  - REST API Development (FastAPI/Flask).
  - Database Schema Implementation (SQLAlchemy).
  - Notification System (Email Alerting Service).
- **5. Presentation Layer (Layer 4)**
  - Web Dashboard UI/UX Development.
  - Real-time Results Visualization (Charts & Gauges).
  - Query History Management & Exporting (PDF/CSV).

3. Resource Management & Roles

3.1 Team Structure & Responsibilities

To achieve the project objectives within the specified timeline, responsibilities are divided among the 6 team members based on their technical expertise. Each member is responsible for a specific layer of the Mini SIEM architecture, ensuring a clear separation of concerns.

Role	Key Focus Area	Primary Responsibility
Project Manager	Leadership & Documentation	Overseeing milestones, managing deadlines, and preparing technical reports (D1, D4).
Detection Engineer (Andrew)	**Security Logic**	Developing the rule-based detection engine and mapping events to the **MITRE ATT&CK** framework.
Backend Developer	API & Integration	Building the REST API gateway, event ingestion pipelines, and notification services.
ML Engineer	Scoring & AI	Designing and training the Machine Learning model for automated threat confidence scoring.
Frontend Developer	User Experience (UX)	Developing the web dashboard, visualising results, and implementing the Help/History pages.
Data Engineer	Connectivity & Persistence	Developing API connectors (VirusTotal, AbuseIPDB, OTX) and managing the SQL database schema.

Export to Sheets

3.2 RACI Matrix

The RACI Matrix clarifies the level of involvement for each member in the project's major work packages. *(R: Responsible, A: Accountable, C: Consulted, I: Informed)*

Task / Work Package	PM	Detection Eng.	Backend Dev.	ML Eng.	Frontend Dev.	Data Eng.
Requirements & Design Docs	**A/R**	C	C	C	C	C
API Connectors (VT, OTX)	I	C	C	I	I	**A/R**
Data Ingestion & Normalization	I	C	**A/R**	C	I	R
ML Scoring Engine	I	C	I	**A/R**	I	C
Rule-Based Detection Logic	I	**A/R**	C	C	I	I
Web Dashboard UI/UX	I	I	C	I	**A/R**	I
Final Documentation & Demo	**A/R**	R	R	R	R	R

4. Project Timeline & Milestones

4.1 Project Phases & Schedule

The Mini SIEM project is executed across five distinct phases over the first half of 2026. Each phase represents a critical step in transitioning from conceptual design to a fully functional security platform. The schedule is designed to allow for parallel development of the detection engine and the machine learning models.

4.2 Gantt Chart (Project Schedule)

Phase	Task / Work Package	Duration	Start Date	End Date
Phase 1: Foundation	Requirements, System Design (UML/DFDs), and Project Planning	5 Days	June 15	June 19
Phase 2: Core Development	API Connectors (VT, OTX) , Data Ingestion , and Backend API	10 Days	June 20	June 29
Phase 3: Intelligence Layer	ML Scoring Engine and Rule-Based Detection Logic	10 Days	June 25	July 4
Phase 4: Frontend & Viz	Web Dashboard UI/UX and Real-time Visualization	7 Days	July 2	July 8
Phase 5: Integration & QA	System Integration, Debugging, and Security Testing	5 Days	July 9	July 13
Phase 6: Final Delivery	Final Documentation (D1), Demo Video (D4), and Submission	4 Days	July 14	July 17

4.3 Key Project Milestones

Milestones are fixed points in the project timeline that signify the completion of a major deliverable.

Milestone	Deliverable Description	Target Date	Status
M1: Project Charter	Finalization of Requirements and System Design.	Feb 2026	Completed
M2: Backend Ready	API Gateway and Ingestion Layer operational.	March 2026	Completed
M3: Intelligence Core	ML Scoring and Detection Engine integration.	April 2026	Completed
M4: Full Integration	Web Dashboard connected to the live backend.	May 2026	In Progress
M5: Final Delivery	Submission of code, documentation, and demo.	May 2026	Upcoming

5. Risk Management Plan

5.1 Risk Identification & Assessment

A proactive risk management approach is essential for the Mini SIEM project to anticipate potential obstacles and develop effective mitigation strategies. Risks are categorized into Technical, Operational, and Schedule-based risks.

5.2 Risk Mitigation Matrix

The following table outlines the primary risks identified for this project and the corresponding plans to minimize their impact.

Risk Category	Description	Impact	Mitigation Strategy
Technical	**API Rate Limiting:** Exceeding the free tier limits of VirusTotal or AbuseIPDB.	High	Implement local caching for frequent queries and optimize API calls to avoid redundant requests.
Technical	**Inaccurate Detection:** The rule engine producing too many "False Positives" due to poor logic.	High	**(Andrew's Focus):** Rigorous testing using known malicious and benign samples to fine-tune the rule-based logic.
Operational	**Data Inconsistency:** Different APIs returning conflicting information for the same IOC.	Medium	Develop a "Smart Scoring" algorithm that assigns weights to different sources to produce a unified verdict.
Project	**Schedule Slippage:** Delays in the ML model training phase affecting the final integration.	Medium	Utilize pre-trained models for initial testing and allocate "buffer days" before the final May 2026 deadline.
Resource	**Integration Conflicts:** Issues when merging code from all 6 team members on GitHub.	Medium	Enforce strict version control practices, including frequent commits and peer code reviews before merging.

5.3 Quality Assurance (QA) Procedures

To ensure the project meets the required standards, the following QA measures are implemented:

- **Unit Testing:** Each API connector is tested independently.
- **Security Testing:** Sanitizing all inputs to prevent SQL Injection or Cross-Site Scripting (XSS).
- **User Acceptance:** Conducting trial queries with "dummy" data to verify the Dashboard visualization.

6. Communication & Project Closing

6.1 Communication Management Plan

To ensure seamless collaboration and transparency, Group 6 utilizes a structured communication strategy. This ensures that technical blockers are identified early and all members remain aligned with the May 2026 deadline.

Communication Channel	Frequency	Primary Purpose
Daily Sync-up	Daily (Online)	Quick status updates on individual tasks and immediate blockers.
Weekly Progress Review	Every Saturday	In-depth review of the week's milestones and technical adjustments.
GitHub Repository	Continuous	Version control, peer code reviews, and issue tracking.
Technical Documentation	Ongoing	Updating the project wiki and ensuring code comments are clear.
Instant Messaging	Real-time	Urgent coordination and informal team communication.

6.2 Final Project Deliverables

The following deliverables represent the successful completion of the Mini SIEM project and will be submitted for the final evaluation:

1. **Technical Documentation:** Comprehensive Project Plan, System Design, and API Documentation.
2. **Functional Web Prototype:** The live Mini SIEM application featuring the detection engine and dashboard.
3. **Source Code:** Fully documented Python and JavaScript code hosted on GitHub.
4. **Demonstration Video:** A narrated walkthrough highlighting the IOC analysis and ML-scoring features.
5. **Project Presentation:** A final slide deck summarizing the project's impact and architecture.

6.3 Conclusion & Future Enhancements

The Mini SIEM project successfully integrates threat intelligence with machine learning to provide a streamlined analysis tool for security analysts. While the current prototype meets all core objectives, the team has identified future enhancements, including:

- **Active Remediation:** Integrating with firewalls to automatically block malicious IPs.
- **Advanced Visualization:** Implementing 3D threat mapping for global IOC trends.
- **Expansion:** Adding support for log file analysis from Windows Event Logs and Linux Syslog.